

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EE)

Nov 2013

EE304.01 Industrial Instrumentation

Date: 20.11.2013, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Answer the following:

[07]

- 1) State the name of 2 static characteristics in the measurement systems
- 2) A pressure measurement instrument is calibrated between 10bar and 250 bar. The scale range of the instrument is _____.
- 3) A diaphragm type pressure measuring instrument is calibrated for absolute pressures of 6 to 760mm of mercury. It has an accuracy of $\pm 1\%$ based upon scale span. Calculate the scale span and maximum static error.
- 4) The true value of voltage across a resistor is 50V. The measurement finds a value of 49V. Calculate absolute error, percent error and percent accuracy.
- 5) A flow meter is calibrated from 0 to 100 m³/s. The accuracy is specified as within $\pm 0.75\%$ above 20% of scale reading. What is the static error if the instrument indicates 80m³/s?
- 6) State one point of difference between LVDT and RVDT
- 7) State the only elastic type pressure measurement device used for absolute as well as differential pressure measurement

Q - 2 (a) A bimetallic thermometer is made up of strips of a nickel chromium alloy and Invar bonded together at 25°C. Each strip has a thickness of 1mm and a length of 50mm. Calculate the radius of curvature produced when the strip is unstrained and subjected to a temperature of 200°C. For nickel chrome alloy α is $12.5 \times 10^{-6} / ^\circ\text{C}$ and Invar α is $1.7 \times 10^{-6} / ^\circ\text{C}$

[04]

(b) A strain gauge having an unstrained resistance of 350 Ω and a gauge of 2 is connected in series with a ballast resistance across a 10V supply. The ballast resistance is designed to give maximum sensitivity. The gauge is subjected to a dynamic strain of $(10 + 20 \sin 314 t) \mu\text{strain}$.

[04]

- a) Find the expression for the change in output voltage on account of strain
 - b) If a capacitor is connected in one output leads and if the voltage is read by a true rms reading voltmeter of infinite impedance, find the output voltage.
- (c) Explain construction, working, advantages, disadvantages and range of a low pressure measuring technique which uses thermal conductivity as its principle of measurement

[06]

OR

Q - 2 (a) List out various types of proximity sensor. Explain principle, construction, working, advantages and disadvantages of both types of proximity sensor. Also state one application of any one of the proximity sensor

[07]

(b) Explain in detail the principle, construction, working, advantages, disadvantages and range of IR type gas analyser

[07]

Q - 3 (a) List out various techniques for dew point measurement and explain principle.

[04]

construction and working of any one technique.

- (b) List out various method for torque measurement and explain principle, construction and working of any one technique. [05]
(c) Derive the equation of gauge factor. [05]

OR

- Q - 3 (a) List out various differential pressure flow measurement techniques and explain principle, construction, working, advantages and disadvantages of any one of the technique [07]
(b) Explain construction and working of thermister [03]
(c) Explain construction and working of piezoelectric transducer for measurement of vibration. [04]

SECTION - II

- Q - 4 (a) Match the following [07]

- | | | | |
|---|--------------------------|----|------------------|
| 1 | Bimetallic sensor | a) | Turbine |
| 2 | Direct level measurement | b) | Ionization gauge |
| 3 | Mass flow meter | c) | Digital Action |
| 4 | Pressure measurement | d) | Cell Constant |
| 5 | Active transducer | e) | Sight glass |
| 6 | Conductivity | f) | Recorder |
| 7 | Data Storage | g) | Piezoelectric |

- Q - 5 (a) Multiple choice questions:

[14]

- 1) A pressure gauge is calibrated from 0-50kN/m². It has a uniform scale with 100 scale divisions. One fifth of a scale division can be read with certainty. The gauge has a:
 - a) Resolution of 0.1kN/m²
 - b) Threshold of 0.1 kN/m²
 - c) Dead zone of 0.2 kN/m²
 - d) Resolution of 0.5 kN/m²
- 2) The measurement of a quantity
 - a) Is an act of comparison of an unknown quantity with another quantity
 - b) Is an act of comparison of an unknown quantity with a known quantity whose accuracy may be known or may not be known
 - c) Is an act of comparison of an unknown quantity with a predefined acceptable standard which is accurately known
 - d) None of the above
- 3) A transducer converts
 - a) Mechanical energy into electrical energy
 - b) Mechanical displacement into electrical signal
 - c) One form of energy into another form of energy
 - d) Electrical energy into mechanical form
- 4) Moire fringe are used along for measurement of rotary displacement along with
 - a) Only contact type encoders
 - b) Only optical encoders
 - c) Both contact type and optical encoders
 - d) None of the above

- 5) In nuclear instrumentation which are the heaviest and slow moving particles?
 - a) Beta particles
 - b) Gama particles
 - c) Alpha particles
 - d) Neutrons
- 6) Law of intermediate metals in thermocouples allows them to:
 - a) Use reference junction compensation
 - b) Use meters for measurement without disturbing the circuit conditions
 - c) Use of extension wires of materials other than the one for making thermocouples
 - d) Both b) and c)
- 7) Pirani gauge is used for measurement of pressure
 - a) Between 10^{-1} to 10^{-3}
 - b) 10^{-1} to 10^{-6} mm of Hg
 - c) 10^{-1} to 10^{-9} of Hg
 - d) 10^{-1} to 10^{-12} mm of Hg
- 8) Dummy strain gauges are used for:
 - a) Compensation of temperature changes
 - b) Increasing the sensitivity of bridge in which they are included
 - c) Compensating for different expansion
 - d) Calibration of strain gauges
- 9) In dc tachogenerators used for measurement of speed of shaft, frequent calibration has to be done because
 - a) The contact wear off
 - b) The strength of permanent magnet decreases with age
 - c) The armature current produces heating effects
 - d) All of the above
- 10) A 0-300V voltmeter has an error of $\pm 2\%$ of full scale deflection. What would be the range of readings if true voltage is 30V?
 - a) 24V-36V
 - b) 29.4V- 30.6
 - c) 20V-40V
 - d) None of the above
- 11) A set of readings has a wide range and therefore it has
 - a) Low precision
 - b) High precision
 - c) Low accuracy
 - d) High accuracy
- 12) A null type instrument as compared to deflection type instrument has
 - a) Higher accuracy
 - b) Lower sensitivity
 - c) Faster response
 - d) All of the above
- 13) Which of the instruments mentioned below are used for CO measurement
 - a) Thermal conductivity gas analyzer
 - b) Conductivity meter
 - c) pH meter
 - d) Electromagnetic flow meter
- 14) For measuring a temperature range of 500°C which sensor can be used
 - a) RTD

- b) K_t type Thermocouple
- c) Radiation pyrometer
- d) Bimetallic strip

OR

Q - 5 (a) Attempt any two:

[14]

- 1) X-Y Recorder
- 2) Strip Chart Recorder
- 3) Digital Tape Recorder

Q - 6 (a) Explain in detail the steps you will follow to design an electromagnetic flow meter. Assume necessary data and show the calculation where required. [10]

(b) Explain the construction and working of calomel electrode [04]

OR

Q - 6 (a) Explain the principle, construction, working, advantages, disadvantages and range of radiation type level sensing technique. Also explain the construction and working of three radiation detectors [14]
